
Long-range Dependency is all you need in sEMG Signal Processing

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Abstract

We evaluated CNN, RNN, and hybrid architectures derived from the TDSCNN baseline and the alternative ConvResNet on single-user sEMG data. Training data were augmented with down-up noise injection and temporal cropping augmentation to mitigate overfitting. Owing to their ability to capture long-range dependencies, RNN-based models outperformed CNNs, with our best model achieving a test CER of 11.80%, a 10.37% absolute reduction relative to the baseline.

1 Introduction

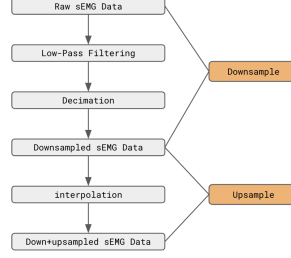
Our primary objective is to design a model pipeline that minimizes the testing CER on single-user sEMG data. We aimed to understand the differential effectiveness of CNNs and RNNs in capturing long-range dependencies. Furthermore, we were interested in advanced data preprocessing and augmentation strategies—applied to both raw sEMG signals and their log-spectrogram representations—to enhance model performance while mitigating overfitting.

2 Method

2.1 Data Augmentation and Data Preprocessing

We implemented **Downsampling-Upsampling Noise Injection** for **Data Augmentation**, by first downsampling the raw sEMG training dataset by applying decimation and then upsampled it back to the original length by using interpolation. Then, we combined the original training dataset with the Down-Up noisy data to obtain the augmented training dataset (Down-Up Boosted Dataset). Models were trained on the augmented training dataset but still validated and tested on original validation and testing datasets. In the downsampling step, the signal is first low-pass filtered to prevent aliasing, and then reduced in temporal resolution by keeping every n^{th} sample (where n is the downsampling factor). This operation decreases the sampling rate (e.g., from 2000 Hz to 1000 Hz for $n = 2$) while the spatial dimensions (e.g., electrode channels) remain unchanged. Next, the downsampled signal is then upsampled via interpolation to restore the original number of samples. The Downsampling-Upsampling mechanism is demonstrated in the Figure 1.

Figure 1: Downsampling-upsampling architecture



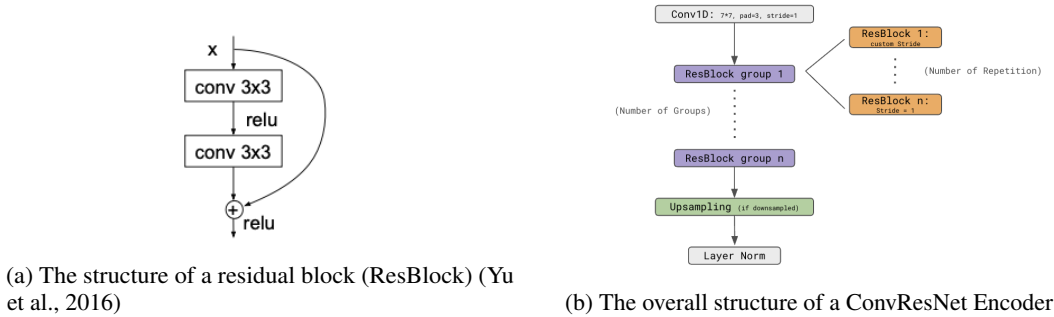
Observing no significant effect upon data preprocessing schemas on the frequency domain, we speculate that sole preprocessing on the temporal domain might render more robust results. Accordingly, we experimented with applying **Temporal Random Crop to the Log-spectrogram** for **Data Preprocessing**, which randomly selects and removes a continuous temporal segment along the time dimension. The function takes an argument specifying the number of frames to be removed and crops a maximum time segment of $\frac{1}{125} \times \text{crop_size} \times 1000$ ms. The cropped segment is then resized back to the original time length using linear interpolation in PyTorch. Interpolation works by estimating the missing values between known points using a weighted average for each frequency bin, effectively smoothing transitions and preserving temporal structure. It ensures that the time dimension is restored without abrupt discontinuities, allowing the model to process a coherent spectrogram.

2.2 Architecture

2.2.1 An alternative CNN: ConvResNet

Taking inspiration from the general success of TDSConv as a parameter-efficient and Convolutional Neural Network architecture specialized for ASR-like tasks, we first attempted alternative CNN structures like **ConvResNet**. Wang et al. in their 2017 work identified the potential of such structure in Automatic Speech Recognition optimized by CTC loss [8]. Based on their architecture, we made slight modifications and attempted multiple parameter setups. The architecture features the critical **ResBlock** which contains two identical convolutional layers with 3×3 filter, with the skip connection being the identity mapping of input x . In our design of the **ConvResNet Encoder**, we generally adopt the model pipeline of Wang et al., 2017 by pipelining ResBlock Groups each contains certain repetitions of ResBlocks with the first one taking a custom stride and the rest taking a fixed $\text{stride} = 1$ [9]. To account for the downsampling effect brought by $\text{stride} > 1$, a learnable upsampling module to restore the original sampling rate was inserted before the layer norm. The ConvResNet Encoder architecture is given by Figure 2.

Figure 2: Architectural design of the ConvResNet Encoder



2.2.2 CNN + RNN: From ConvResNet+GRU to TDSConv+LSTM

The unsatisfying performance of a sole **ConvResNet Encoder** sparked new directions of architectural design. Upon the scaling of performance along less restraints on the model's learning, such as

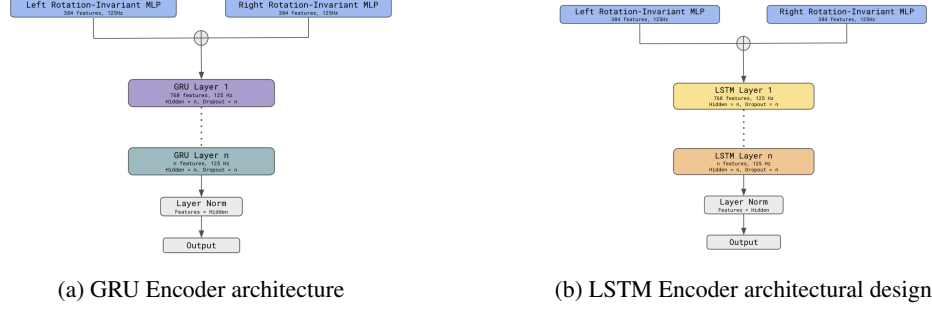
canceling bottleneck, lowering downsampling rate which on the other hand imposes the drawback of limiting the receptive field on the temporal dimension, we reckon that capturing long-range dependency information might aide the performance of the encoder. With that intention, we attached a GRU layer atop the ConvResNet taking the output of the ConvResNet module as its input and outputting features in the dimension of `hidden_dim`, to build a hybrid **ConvResNet+Bi-GRU Encoder**.

Figure 3: Architectural design of the CNN+RNN Hybrid Encoders

2.2.3 The power of RNN: a comparative approach towards LSTM and GRU

We first attempted to install a Gated Recurrent Unit (GRU) encoder in the place of the TD-Conv encoder. The new **GRU Encoder** essentially features a stack of bidirectional GRU layers. Each layer is equipped with hidden and cell states that update over each timestep. Parameters for the encoder are number of hidden dimensions, number of layers, and the dropout rate. By tuning the parameters `hidden_dim`, `num_layer`, and `dropout`, we were able to render GRU encoders of varying complexity.

Figure 4: Architectural design of RNN Encoders: GRU and LSTM



2.3 Number of Electrode Channel Exploration

We also explore how many channels of electrodes are enough to efficiently achieve strong decoding performance. A function was implemented at the preprocessing stage, controlled by three hyperparameters: the **start** index of the selected channel (inclusive), the **end** index of the selected channel, and the number of channels on each side to **pick out selected channels data among all the channels**. Regarding our limited understanding for the information provided by each channel, the selection always starts from index 0 to the specified channel index. Experiments were conducted over 20 epochs on the baseline model to compare variations in performance with the number of channels.

2.4 Amount of Training Data Exploration

To investigate how much training data is required to maintain strong decoding performance, we **randomly subset the dataset before training**. The proportion of data retained is controlled by an argument passed to the subsetting function. We ran experiments for both 20 and 50 epochs to illustrate the effects of different random subgroups on the baseline model's performance.

3 Results

3.1 Data Augmentation and Data Preprocessing

3.1.1 Data Augmentation: Downsampling-Upsampling Noise Injection Raw Data Boosting

We trained the baseline model on augmented training dataset using the **Downsampling-Upsampling** mechanism with **factor=2** (2000Hz $\rightarrow$ 1000Hz $\rightarrow$ 2000Hz) and **factor=4** (2000Hz $\rightarrow$ 500Hz $\rightarrow$ 2000Hz). Models are trained on 20 epochs to investigate the trend between values of factor and CER. Training and testing results are shown in the table 1.

	Down-up Factor	Train CER (%)	Val CER (%)	Test CER (%)
<i>Original Data</i>	None	70.34	65.88	64.30
<i>Down up Boosted Data-1</i>	4	77.69	64.73	63.13
<i>Down up Boosted Data-2</i>	2	49.09	36.82	34.93

As expected, the augmentation introduces controlled temporal distortions in the sEMG signal, simulating variations in sensor resolution and forcing the model to learn robust and invariant features [5]. We believe that this augmentation technique injects more noise into the total training data set, when combined with the original data. This strategy has been previously shown to improve generalization performance in sequence modeling tasks [5]. This augmentation is expected to introduce **discrete temporal distortion** upon sampled data points from the signal by taking out certain portion of ground-truth information and injecting noise in a controlled manner. This strategy would effectively boost the size of the training data set when combined with the original data set.

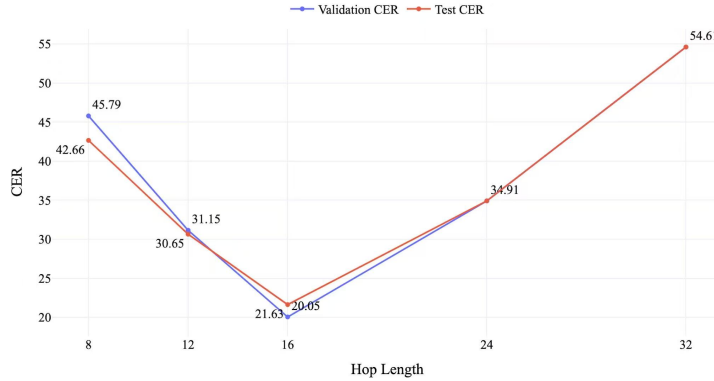
The table above compares the performance of baseline model trained on original training dataset and Down-Up boosted training dataset, clearly showing the increase in performance and decrease in overfitting of Down-Up boosted training dataset. Moreover, Down-Up Boosted Data-2 with smaller down-up factor preserves more temporal detail in the sEMG signal after downsampling and upsampling, which in turn yields lower CER values across training, validation, and testing.

3.1.2 Data Preprocessing: Temporal Random Cropping and Raw Data Downsampling

The Temporal Random Crop technique resembles the data augmentation strategies initially incorporated into the baseline model, but it specifically targets the time dimension. By randomly removing a segment and interpolating it back to the original length, we ensured that the frequency information remained unchanged while modifying the temporal structure. As anticipated, providing the model access to more diverse temporal patterns led to a significant improvement in performance. With the same 200ms time mask, the model achieved a validation CER of 61.41, compared to the baseline 65.88 after 20 epochs of training. We believe that this preprocessing schema simulates introducing **continuous temporal distortion** that is projected to help the model learn robust features on the long run, thus capturing the underlying pattern. The improved performance with sole manipulation on temporal domain instead of both temporal and frequency domains also highlights the significance of frequency continuity and completeness in speech recognition models as preservation of the information on the frequency domain helps maintain stable performance.

To investigate the relationship between sampling rate and CER, we experimented with `hop_length` parameter during the log spectrogram transformation. `hop_length` parameter determines the step size between consecutive STFT windows and serves as a downsampling factor for the raw signal. We trained and tested the baseline model with various `hop_length` values, which correspond to different raw data sampling frequencies. By the results shown in the plot in Figure 5, we found out `hop_length=16` ($125Hz$) generated best performance, which is already used by the baseline model. When `hop_length<16`, the spectrogram frames are overly dense, capturing excessive micro-variations and risking overfitting [5]. Conversely, `hop_length>16` discards key transient features [5]. Hence, `hop_length=16` balances detail retention and overfitting control, yielding optimal CER.

Figure 5: `hop_length` vs. CER(Validation and Test)



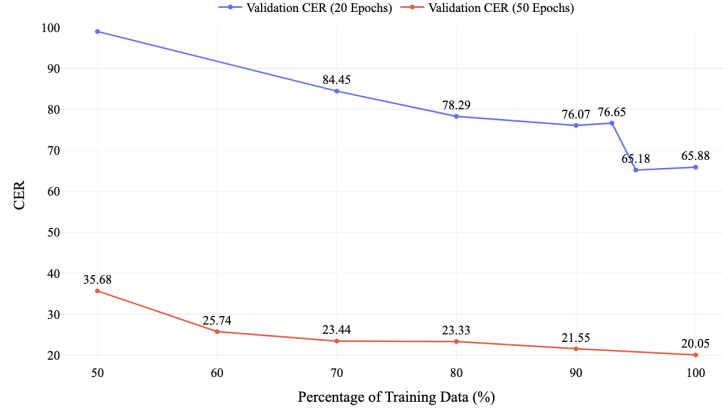
3.2 Data Efficiency

The amount of data required to achieve optimal model learning performance is closely tied to the number of training epochs. When training has limited epochs, having a sufficiently large and complete dataset is crucial for effective learning. We observed that when less than 95% of the data was used, the model struggled to perform, yielding suboptimal results compared to the baseline. However, as the dataset size approached 95% of the original data, there was a sudden boost in performance, reaching levels comparable to the optimal configuration.

In contrast, during longer training sessions, such as 50 epochs, the model relies less on having the full dataset, as extended training allows it to extract meaningful patterns even with a reduced amount of data. In our experiments, we found that using only 60% of the original dataset was sufficient

to achieve comparable performance to training on the full dataset. This indicates that while short training durations demand more data for rapid learning, longer training mitigates the impact of data reduction by allowing the model to refine its understanding over time. The statistics are demonstrated in the plots of Figure 6.

Figure 6: Validation CER for Different Amount of Training Data



3.3 Channel Efficiency

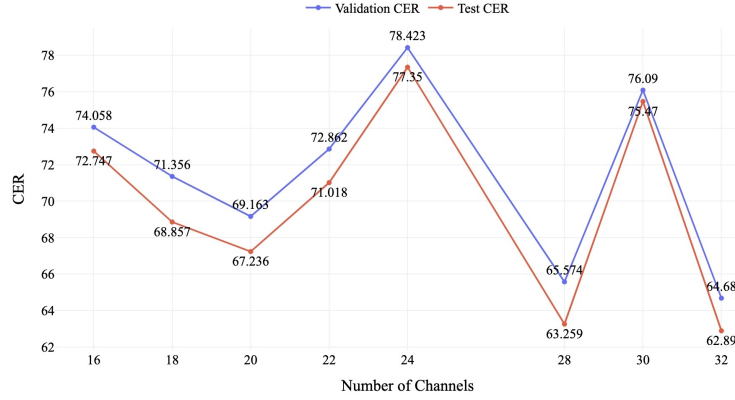
The relationship between the number of channels and the baseline model’s performance is non-linear. The first notable improvement in Test CER occurred with 20 channels, which required only 62.5% of the original channels while still achieving results comparable to the baseline. When training was extended to 50 epochs, this 20-channel configuration yielded validation CERs of 26.18 and 28.05, figures very close to the baseline model’s performance. Additionally, a configuration with 28 channels produced performance nearly identical to using all channels, suggesting that a full set of channels is not strictly necessary to achieve optimal performance.

We also observed significant fluctuations in validation CER as the channel count increased, leading us to suspect potential collinearity among adjacent channels. This suggests that certain channels may share or depend on each other’s information for effective learning. For example, when comparing the performance of using the first 10 channels (out of 20 total) to using channels 11 – 13, we observed a higher CER in validation and testing with the latter. Specifically, including channels 11 – 13 led to worse performance compare to only include 10 channels. This is likely because the model struggled without channel 14, which was crucial for supporting the learning information from channels 11 – 13.

A similar pattern emerged with channel 15: while the performance of the first 14 channels was similar to using all channels, adding channel 15 in isolation caused confusion and increased CER. This issue was resolved when channel 16 was introduced, further supporting our hypothesis about collinearity among adjacent channels.

Given these dependencies, along with the near-baseline performance achieved with 28 channels, using 28 channels (87.5% of the total) represents a strong compromise between accuracy and computational efficiency. The statistics are demonstrated in the plots of Figure 7.

Figure 7: Validation and Test CER for Different Number of Channels



3.4 Architectural Experiment

3.4.1 The battle of CNN architecture: TDSCov vs. ConvResNet

ConvResNet

We tested multiple parameter settings including different group numbers, repetition numbers, and stride size, yet achieving poor performance given by Table 2, compared with the Baseline model performance achieving a Test CER of mere 21.63 under the same 50-epoch training setting. The significantly higher test CER suggests that the ConvResNet model is comparatively hard to train and is not projected to well fit the data after 150 epochs with the single-user data. Accordingly, for the purpose of our study, we may believe that TDSCov is a better CNN model regarding our experimental setting and objectives.

Table 2: Testing statistics of the ConvResNet Encoder under different num_group configurations, 50 epochs

	Groups	Dimensions	Repetitions	Strides	Bottleneck	Test CER (%)
<i>ConvRes-1</i>	4	[64, 96, 128, 256]	[3, 2, 2, 2]	[1, 2, 2, 2]	[false, false, true, true]	92.69
<i>ConvRes-2</i>	3	[64, 128, 256]	[3, 2, 2]	[1, 1, 2]	[false, false, false]	68.66

3.4.2 The redundancy of CNN + RNN architecture

ConvResNet + GRU

Compared with the previous ConvResNet Encoder, the result was shockingly good: a drastic decline in Test CER of more than 40% is observed with dropout = 0.0 as shown in Table 3.

Table 3: Testing statistics of the ConvResNet + GRU Encoder of different GRU hidden_dim configurations, 50 epochs

	Groups	Dimensions	Repetitions	Strides	GRU Hidden	Test CER (%)
<i>ConvRes+GRU-1</i>	3	[64, 128, 256]	[3, 2, 2]	[1, 1, 1]	768	20.27
<i>ConvRes+GRU-2</i>	3	[64, 128, 256]	[3, 2, 2]	[1, 1, 1]	512	19.51

With the ConvResNet+Bi-GRU Encoder, we could match the baseline performance after 50 epochs of training. Since we have already discovered that TDSCov outperforms ConvResNet in the experimental setting of this study in all regards, we transition to establish the TDSCov+Bi-LSTM Encoder to leverage TDSCov’s performance and parameter-efficiency, meanwhile taking advantage of LSTM’s finer long-range dependency details.

TDSConv + LSTM

We indeed rendered results of what we have expected: the TDSConv+BiLSTM is able to achieve a slightly better testing CER of 18.60 at 50 epochs generally (compared with the best testing CER of ConvRes+GRU 19.51). But another issue arises: The model is already heavily overfitting with mere 50 epochs. As we aim to train the data with more generalizability while at the same time maintaining high performance on the single-user case, we tried different regularizer strategies with results shown in Table 4.

Table 4: Testing statistics of the TDSConv + LSTM Encoder of different LSTM configurations, 50 epochs

	LSTM Hidden	LSTM Dropout	Train CER (%)	Val CER (%)	Test CER (%)
<i>TDSConv+LSTM-1</i>	512	0.1	9.49	17.89	18.60
<i>TDSConv+LSTM-2</i>	256	0.2	14.16	18.20	20.40

Nonetheless, the results are not satisfying as simplifying and regularizing the LSTM did not seem to achieve the expected effect: we are observing a training CER of 14.16 which is more than 6% lower than the corresponding testing CER in the best regularized case (*TDSConv+LSTM-2*). Overfitting still appears to be a major problem and stricter regularization would all the performance advantages gained over the baseline implementation. We reflect upon the problem and speculate that this could be the redundancy and the overwhelming complexity brought by the hybridization of CNN and RNN models.

3.4.3 Regulating RNNs to harness the power of long-range dependency

GRU

We attempted the following parameter initializations for the GRU Encoder and respectively got the result shown in Table 5. As we may see that the performance of the GRU encoder generally align with our expectations: *GRU-1* with `hidden_dim = 768` scores an impressive testing CER of 17.19. Nonetheless, we also suffered from heavy overfitting with mere 50 epochs of training when the `hidden_dim` is relatively large. Notably, the best performing *GRU-1* scores a mere 7.69 in training CER which is approximately 10% less than the testing CER, accompanied by a flattened fluctuation in loss, suggesting learning noise instead of groundtruth pattern.

Table 5: Testing statistics of various Bi-GRU Encoders, 50 epochs

	Hidden	Layers	Dropout	Train CER (%)	Val CER (%)	Test CER (%)
<i>GRU-1</i>	768	3	0.0	7.69	24.52	17.19
<i>GRU-2</i>	512	3	0.0	9.19	23.21	21.53
<i>GRU-3</i>	256	3	0.2	18.71	19.52	19.92

Deploying the classic `drouput = 0.2`, we are able to contain the fitting of the encoder under the desired range of effective learning with a training CER of 18.71 and a validation CER of 18.20. We referred to the similar regularizing strategy when training the LSTM Encoder, the detailed process of which is specified in the following discussion.

LSTM

We initialized the parameters with `hidden_dim=512`, `num_layer=3`, `bidirectional=True`, and `dropout_rate=0.0`. We trained it for 150 epochs and achieved an exceptional validation CER of 12.76 (Baseline model, 150 epochs, validation CER=18.91) and a testing CER of 14.71 (Baseline model, 150 epochs, testing CER=22.17). However, the model’s training CER was 2.10, clearly indicating that the model was again overfitting. We first tuned parameter `num_layer`.

Based on the loss plot in Figure 8 of the previous training on 150 epochs, we found that around 50 epochs (6350 batches), the loss started to decrease slowly. Then, we decided to tune number of hidden dimension on 50 epochs for `hidden_dim=512` (*LSTM-1*) and `hidden_dim=256` (*LSTM-2*).

Results are shown in the Table 6 *LSTM-2* with `hidden_dim=256` was less overfitting. As expected, with a smaller hidden dimensionality (256 instead of 512), the *LSTM-2* has fewer parameters, which reduces its capacity to memorize the training data and thus leads to less overfitting.

Figure 8: Loss of LSTM when `hidden_dim=512`, 150 epochs: plateaued after 50 epochs

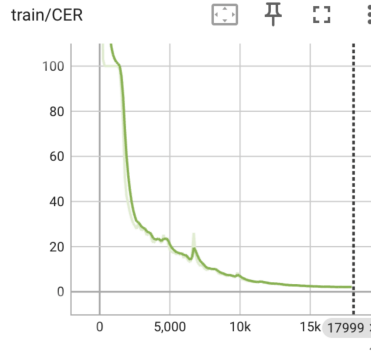


Table 6: Testing statistics of Bi-LSTM encoder for different `hidden_dim`, 50 epochs

	Hidden	Layers	Dropout	Train CER (%)	Val CER (%)	Test CER (%)
<i>LSTM-1</i>	512	3	0	7.46	15.57	15.93
<i>LSTM-2</i>	256	3	0	12.18	17.37	17.55

We trained the model for 100 epochs to magnify the overfitting phenomenon. We set `hidden_dim=256` and subsequently tuned `num_layers` and `dropout`. As the *LSTM-2* with `hidden_dim=256` still exhibited signs of overfitting, we selected `dropout=0.2` to introduce a regularization effect. We then compared different values of `num_layers`; the results are presented in the Table 7.

Table 7: Testing statistics of Bi-LSTM encoder for different `num_layer`, 100 epochs

	Hidden	Layers	Dropout	Train CER (%)	Val CER (%)	Test CER (%)
<i>LSTM-3</i>	256	3	0.2	4.51	13.76	14.46
<i>LSTM-4</i>	256	2	0.2	6.84	15.00	15.97

As expected, *LSTM-4* with `num_layers=2` is slightly less overfitting than *LSTM-3* with `num_layers=3`, but is still overfitting. Since RNN architectures have been shown to better capture the complex and hierarchical temporal dependencies in sequential data, shallow models with very few layers lack sufficient depth to model intricate patterns effectively [2]. Thus, we did not decrease the number of layers to prevent overfitting with the cost of model performance. To further improve model performance while capping the overfitting effect brought by deep RNN encoders without impairing model depth, we introduced data augmentation to address the issue. Down-up augmented training dataset (down-up factor = 4) and temporal random cropping on log-spectrogram (`crop_size=25`) were used to train *LSTM-4*, resulting our best performance model shown in section 3.6 below. `crop_size=25` is a sweet spot between preventing overfitting and preserving temporal dependencies. We also tested with higher `crop_size=40`, but it resulted in slightly worse testing CER because too much temporal dependencies in sequential data were removed, as shown in the Table 8.

Table 8: Testing statistics for random cropping of different crop size, 150 epochs on augmented data

	Crop Size	Train CER (%)	Val CER (%)	Test CER (%)
<i>Crop-1</i>	25	9.87	12.27	11.80
<i>Crop-2</i>	40	11.05	12.65	13.25

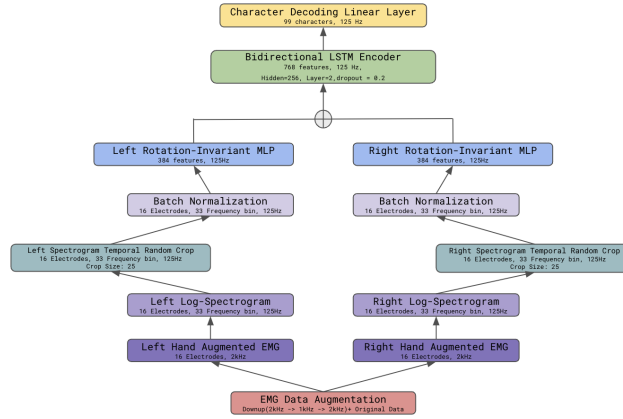
GRU vs. LSTM: Performance Comparison

Overall, we witness the pattern of GRU Encoder’s performance more restricted by harsher regularization schema and its strong propensity towards overfitting at as early as 50 epochs in training. At 50 epochs, we see better potential of scoring lower testing CER capturing LSTM Encoders even when to certain extent overfitting the data, suggesting that LSTM could capture better long-range temporal dependencies than GRU. We thus decided to continue with a fine-tuned LSTM Encoder in the Best-Performance model pipeline and opt to counterbalance its overfitting with data augmentation and data preprocessing.

3.5 Best Performance

The architecture of our Best Performance Model Pipeline is demonstrated in Figure 9.

Figure 9: Overall Architectural Pipeline of the best-performance Model Configuration



Key modules and parameters of our architectural design:

1. Downsampling-Upsampling augmented training dataset, factor = 2;
2. Temporal Random Cropping of Log-Spectrogram, crop_size=25;
3. Bidirectional LSTM Encoder, hidden_dim=256, num_layers=2, dropout=0.2.

Table 9: Comparison of Best Testing statistics of Baseline Model Architecture and Best-performance

	Epochs	Train CER (%)	Val CER (%)	Test CER (%)
<i>Baseline Model</i>	150	2.41	18.91	22.17
<i>Best-performance Model</i>	150	9.87	12.27	11.80

4 Discussion

4.1 Methods not applicable to the task

4.1.1 Data Preprocessing

Contrary to expectations, neither bandpass filtering [1] nor the mel-spectrogram transformation yielded performance improvements. This suggests that though might de facto be a dominant strategy in Automatic Speech Recognition(ASR), log-spacing on frequency domain in signal transformation might not be the most suitable weapon in the arsenal for sEMG signal processing: information preserved by linear spacing on the frequency domain are significant and susceptible to distortion.

Similarly, applying the Hilbert envelope [6] with z-score normalization resulted in a training loss plateau, indicating that the model failed to converge effectively. Interestingly, when we omitted the z-score normalization step, we observed a slight improvement in validation performance, suggesting that preserving raw frequency amplitude differences might retain more relevant signal characteristics. However, even with this minor improvement, the overall performance remained suboptimal. Besides, solely downsampling the training data will not yield ideal performance because temporal resolution and critical transient features are necessary for accurate signal interpretation [5].

4.2 Architectural Design

CNN might have better performance in zero-shot generalization; however, in a single-user setting, it fails to outperform RNN models. We speculate that differences in modeling assumptions unfavor CNN architectures, as they inherently rely on spatially localized patterns and lack mechanisms to effectively capture long-term temporal dependencies. This limitation likely renders more difficulty in training CNN-based encoders in our experimental setting.

4.3 Insights for Performance Enhancement

4.3.1 Temporal distortion aids more robust learning

Our experiments demonstrate that temporal distortion enhances model robustness to variations in sEMG signals. Techniques like Down-Up noise injection (discrete distortion) and temporal random cropping (continuous distortion) help models adapt to realistic changes, including variations in typing speed or slight sensor inaccuracies. Specifically, Down-Up augmentation introduces controlled noise through sampling rate manipulation, while tuned temporal cropping selectively removes time segments yet maintains essential frequency information. Together, these methods enable models to learn stable representations and become more resilient to input variations.

4.3.2 Long-range dependency dominates performance

Capturing long-range dependencies proved essential for improving model accuracy in sEMG decoding tasks, particularly since specific keystrokes often recur after extended intervals. When comparing different RNN architectures, we found that LSTM, with its sophisticated gating and memory mechanisms, effectively captures these nuanced long-term patterns better than GRU, ultimately leading to superior performance. However, both GRU and LSTM models showed tendencies to overfit on single-user data, emphasizing the necessity of employing data augmentation, data preprocessing and regularization techniques to enhance generalizing flexibility and achieve robust performance.

4.4 Future Directions

Several avenues remain unexplored and hold potential for further improvement:

- **Principal Component Analysis (PCA):** PCA on raw data was not explored in our preprocessing steps. Applying PCA could efficiently identify the minimal essential data needed and potentially enhance training efficiency.
- **Convolutional Downsampling:** Convolutional downsampling might effectively downsample the dataset and preserve the key information in the signal at the same time.

- **Randomized Channel Selection:** Our current method selects electrode channels sequentially, which may neglect critical signal information. A systematic, randomized channel selection approach combined with a grid search could identify more informative channels and better characterize channel collinearity.
- **Fine-tuning Hybrid CNN-RNN Models:** Our exploration of CNN-RNN hybrid architectures, while promising, was limited. Additional fine-tuning and exploration of parameter spaces on the CNN module could further leverage the strengths of these combined models.
- **Transformer-based Architectures:** Transformer-based architectures, either alone or in combination with CNN/RNN modules, have yet to be explored. Given their proficiency at capturing long-range dependencies through self-attention mechanisms, transformer-based encoders could further enhance model performance.

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A Appendix

Figure 10: Downsampling-upsampling architecture

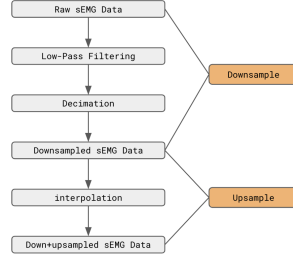
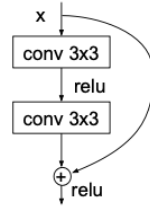
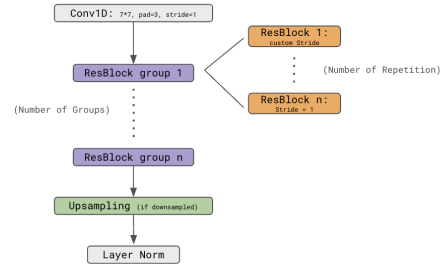


Figure 11: Architectural design of the ConvResNet Encoder

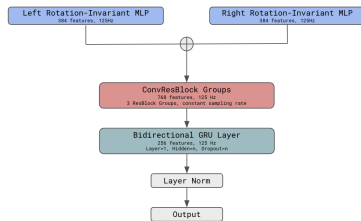


(a) The structure of a residual block (ResBlock) (Yu et al., 2016)

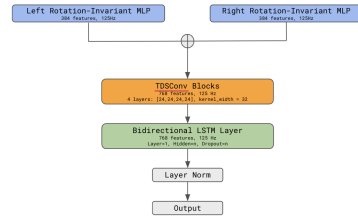


(b) The overall structure of a ConvResNet Encoder

Figure 12: Architectural design of the CNN+RNN Hybrid Encoders: ConvResNet+GRU and TD-SConv+LSTM



(a) ConvResNet+Bi-GRU Encoder



(b) TDSCov+Bi-LSTM Encoder

Figure 13: Architectural design of RNN Encoders: GRU and LSTM

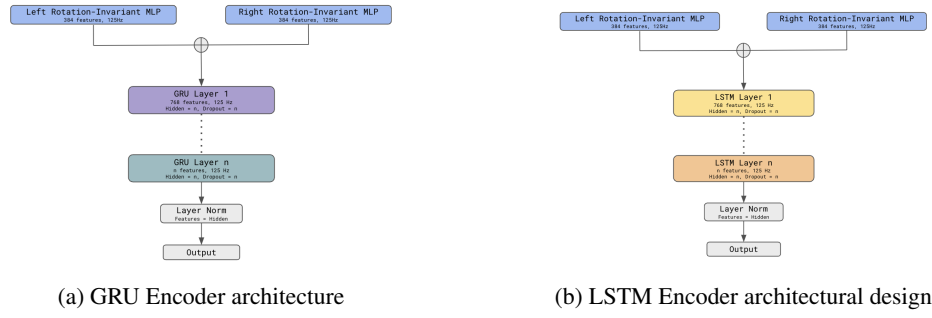


Figure 14: hop_length vs. CER(Validation and Test)

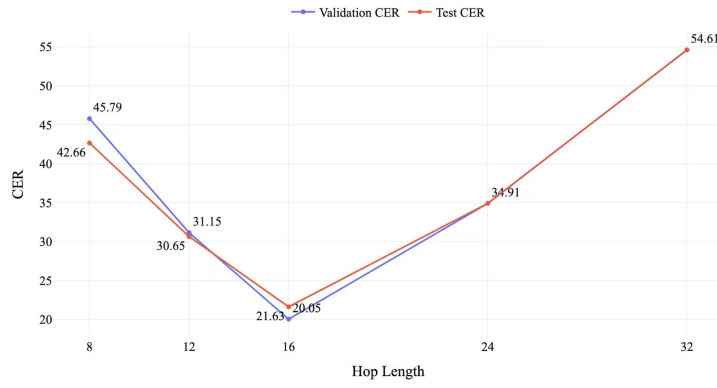


Figure 15: Validation CER for Different Amount of Training Data

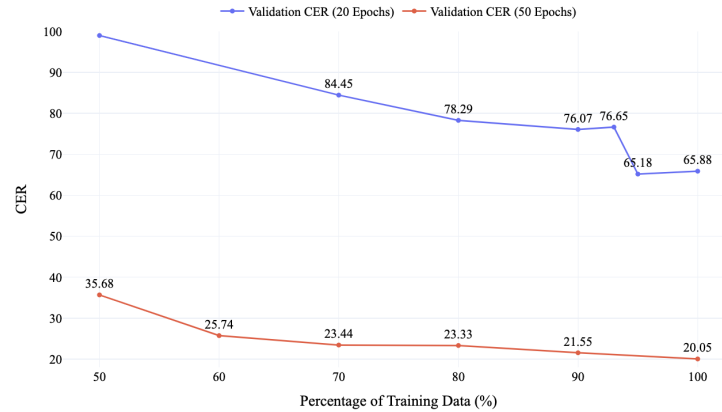


Figure 16: Validation and Test CER for Different Number of Channels

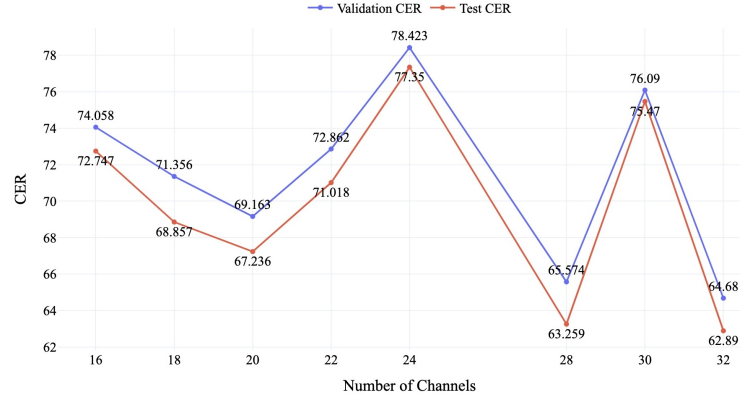


Figure 17: Loss of LSTM when hidden_dim=512, 150 epochs: plateaued after 50 epochs

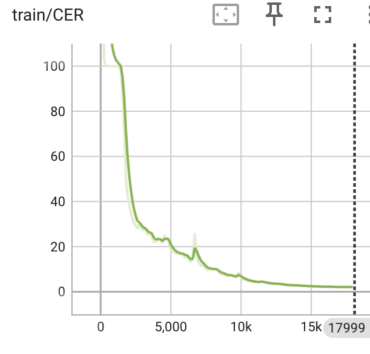


Figure 18: Overall Architectural Pipeline of the best-performance Model Configuration

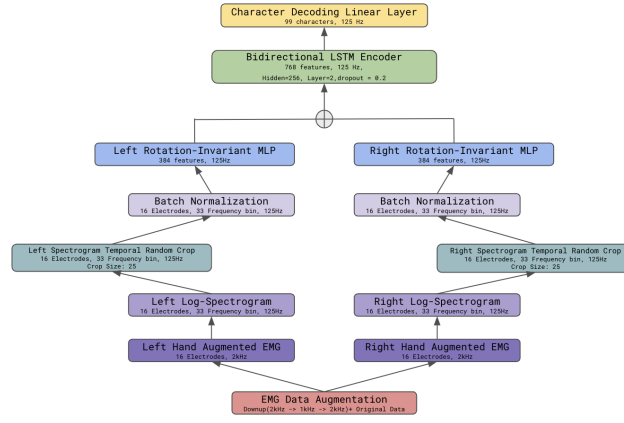


Table 10: Testing statistics for using boosted datasets of different down-up factor, 20 epochs

	Down-up Factor	Train CER (%)	Val CER (%)	Test CER (%)
Original Data	None	70.34	65.88	64.30
Down up Boosted Data-1	4	77.69	64.73	63.13
Down up Boosted Data-2	2	49.09	36.82	34.93

Table 11: Testing statistics of the ConvResNet Encoder under different num_group configurations, 50 epochs

	Groups	Dimensions	Repetitions	Strides	Bottleneck	Test CER (%)
<i>ConvRes-1</i>	4	[64, 96, 128, 256]	[3, 2, 2, 2]	[1, 2, 2, 2]	[false, false, true, true]	92.69
<i>ConvRes-2</i>	3	[64, 128, 256]	[3, 2, 2]	[1, 1, 2]	[false, false, false]	68.66

Table 12: Testing statistics of the ConvResNet + GRU Encoder of different GRU hidden_dim configurations, 50 epochs

	Groups	Dimensions	Repetitions	Strides	GRU Hidden	Test CER (%)
<i>ConvRes+GRU-1</i>	3	[64, 128, 256]	[3, 2, 2]	[1, 1, 1]	768	20.27
<i>ConvRes+GRU-2</i>	3	[64, 128, 256]	[3, 2, 2]	[1, 1, 1]	512	19.51

Table 13: Testing statistics of the TDSCov + LSTM Encoder of different LSTM configurations, 50 epochs

	LSTM Hidden	LSTM Dropout	Train CER (%)	Val CER (%)	Test CER (%)
<i>TDSCov+LSTM-1</i>	512	0.1	9.49	17.89	18.60
<i>TDSCov+LSTM-2</i>	256	0.2	14.16	18.20	20.40

Table 14: Testing statistics of various Bi-GRU Encoders, 50 epochs

	Hidden	Layers	Dropout	Train CER (%)	Val CER (%)	Test CER (%)
<i>GRU-1</i>	768	3	0.0	7.69	24.52	17.19
<i>GRU-2</i>	512	3	0.0	9.19	23.21	21.53
<i>GRU-3</i>	256	3	0.2	18.71	19.52	19.92

Table 15: Testing statistics of Bi-LSTM encoder for different hidden_dim, 50 epochs

	Hidden	Layers	Dropout	Train CER (%)	Val CER (%)	Test CER (%)
<i>LSTM-1</i>	512	3	0	7.46	15.57	15.93
<i>LSTM-2</i>	256	3	0	12.18	17.37	17.55

Table 16: Testing statistics of Bi-LSTM encoder for different num_layer, 100 epochs

	Hidden	Layers	Dropout	Train CER (%)	Val CER (%)	Test CER (%)
<i>LSTM-3</i>	256	3	0.2	4.51	13.76	14.46
<i>LSTM-4</i>	256	2	0.2	6.84	15.00	15.97

Table 17: Testing statistics for random cropping of different crop size, 150 epochs on augmented data

	Crop Size	Train CER (%)	Val CER (%)	Test CER (%)
<i>Crop-1</i>	25	9.87	12.27	11.80
<i>Crop-2</i>	40	11.05	12.65	13.25